

EASTERN INTERNATIONAL UNIVERSITY



MIS 443

Business Data Management Q4

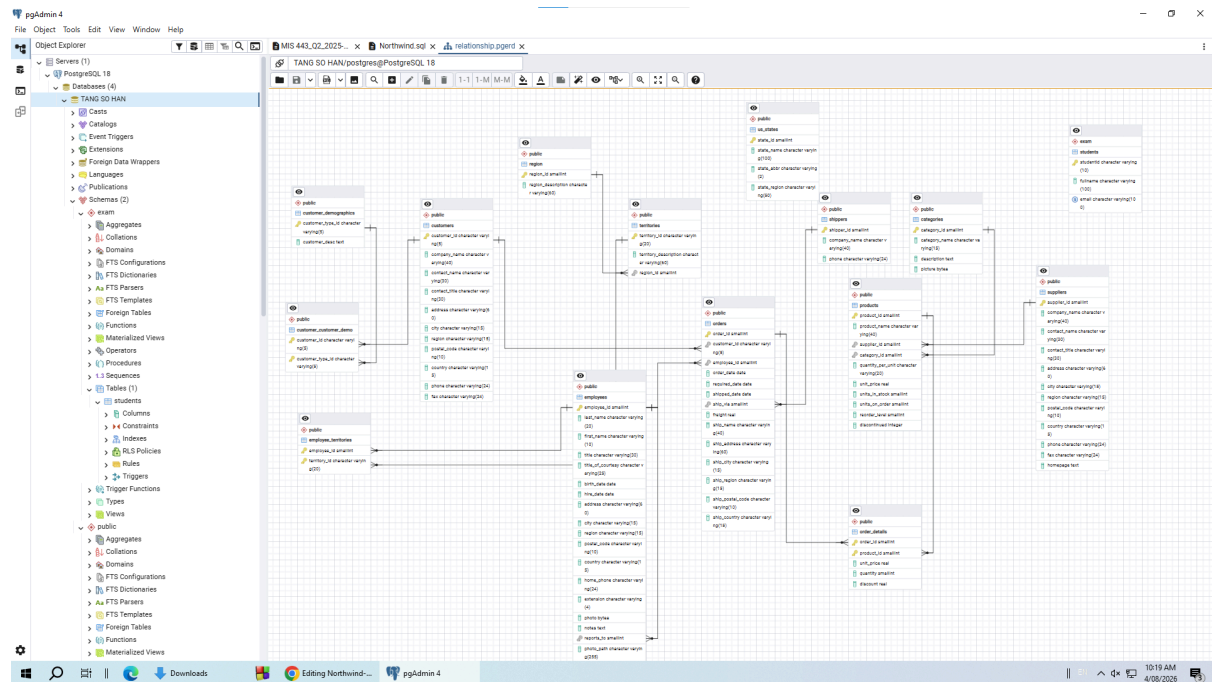
IN - TERM TEST

Word Report

Lecturer: Dang Thai Doan

Student name	Student ID	Github link
Tang So Han	2332300046	https://github.com/hants1825-sys/Northwind-dataset

Table Relationship



Q1

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Publications
- Schemas (2)
 - exam
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (1)
 - students
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (14)
 - categories
 - customer_customer_demo
 - customer_demographics
 - customers
 - employee_territories
 - employees

MS 443_02_2025-2026-Skeleton.sql x Northwind.sql x

TANG SO HAN/postgres@PostgreSQL 18

Query Query History

```
--(a) Must be unique
-- Your answer here
--(b) Create table students
-- Your answer here
create table exam.students (studentid varchar(18) primary key,
fullname character varying(180) not null,
email character varying(180) unique not null);
--(c) Insert your own record
-- Your answer here
INSERT INTO exam.students VALUES ('2332300044', 'Tang So Han', 'han.tang.bbs23@efu.edu.vn');
-- (d) Verify the result
SELECT * FROM exam.students;
```

Data Output Messages Notifications

Showing rows: 1 to 92

table_name	column_name	data_type
name	name	character varying
categories	category_id	smallint
categories	category_name	character varying
categories	description	text
categories	picture	bytea
customer_customer_demo	customer_id	character varying
customer_customer_demo	customer_type_id	character varying
customer_demographics	customer_type_id	character varying
customer_demographics	customer_desc	text
customers	customer_id	character varying
customers	company_name	character varying
customers	contact_name	character varying
customers	contact_title	character varying
customers	address	character varying
customers	city	character varying
customers	region	character varying
customers	postal_code	character varying

Total rows: 92 Query complete 00:00:00.066

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Publications
- Schemas (2)
 - exam
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (1)
 - students
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (14)
 - categories
 - customer_customer_demo
 - customer_demographics
 - customers
 - employee_territories
 - employees

MS 443_02_2025- x Northwind.sql x exam.students/TANG SO HAN/postgres@PostgreSQL 18 x

exam.students/TANG SO HAN/postgres@PostgreSQL 18

Query Query History

```
1 SELECT * FROM exam.students
2 ORDER BY studentid ASC
```

Data Output Messages Notifications

Showing rows: 1 to 1

studentid	fullname	email
PK(character varying(18))	character varying(180)	character varying(180)
2332300044	Tang So Han	han.tang.bbs23@efu.edu...

Servers > PostgreSQL 18 > Databases > TANG SO HAN > Schemas > exam > Tables > students complete 00:00:00.096

pgAdmin 4

Q2

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Publications
- Schemas (2)
 - exam
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (1)
 - students
 - Columns
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - Trigger Functions
 - Types
 - Views
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (14)
 - categories
 - customer_customer_demo
 - customer_demographics
 - customers
 - employee_territories
 - employees

MS 443_Q2_2025-2026-Skeleton.sql x Northwind.sql x

TANG SO HAN/postgres@PostgreSQL 18

Query Query History

```
-- Question 2: Write an SQL query to find the top 5 customers who placed the highest number of orders.
-- Your answer here
select c.customer_id, c.company_name,
count(o.order_id) as total_orders
from customers c
join orders o
on c.customer_id = o.customer_id
group by c.customer_id, c.company_name
order by total_orders desc
limit 5;
```

Data Output Messages Notifications

customer_id	company_name	total_orders
integer	character varying (40)	integer
1	SAVEA	31
2	ERNST	30
3	QUICK	28
4	HUNDO	18
5	FOLKO	18

Showing rows: 1 to 5 Page No: 1 of 1

Total rows: 5 Query complete 00:00:00.077

MS 443 - In-term T... pgAdmin 4

9:58 AM 4/08/2025

Q3

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Publications
- Schemas (2)
 - exam
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (1)
 - students
 - Columns
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - Trigger Functions
 - Types
 - Views
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (14)
 - categories
 - customer_customer_demo
 - customer_demographics
 - customers
 - employee_territories
 - employees

MS 443_Q2_2025-2026-Skeleton.sql x Northwind.sql x

TANG SO HAN/postgres@PostgreSQL 18

Query Query History

```
-- Question 3: Write an SQL query to display a list of orders and the customers who made them. Sort by order date (newest first)
-- Your answer here
select o.order_id, o.order_date, c.customer_id, c.company_name
from orders o
join customers c
on c.customer_id = o.customer_id
order by o.order_date desc;
```

Data Output Messages Notifications

order_id	order_date	customer_id	company_name	
integer	date	integer	character varying (40)	
1	11074	1998-05-05	SMOB	Simone Sistro
2	11077	1998-05-06	RATTC	Rattensville Canyon Grocery
3	11076	1998-05-06	BOLAP	Bon app'
4	11075	1998-05-06	RICRU	Richter Supermarket
5	11071	1998-05-05	LILAS	LILA Supermercado
6	11073	1998-05-05	PERIC	Pericles Comidas clasicas
7	11072	1998-05-05	ERNST	Ernst Handel
8	11070	1998-05-05	LEBMS	Lehmanns Marktstand
9	11068	1998-05-04	QUEEN	Queen Cozinha
10	11069	1998-05-04	TORTU	Tortuga Restaurante
11	11067	1998-05-04	DRACD	Drachenhut Delikatessen
12	11065	1998-05-01	LILAS	LILA Supermercado
13	11064	1998-05-01	VINETC	Vinot Chateau-Moreau
14	11064	1998-05-01	SAVEA	Save-a-lot Markets
15	11061	1998-04-30	GREAL	Great Lakes Food Market
16	11063	1998-04-30	HUNDO	Hungry Out All-Right Diner...
17	11062	1998-04-30	REGGCC	Reggiani Caffe'oli
18	11060	1998-04-30	FRANIS	Franchi S.p.A.
19	11055	1998-04-29	BLAUS	Blaug's Sea Delicatessen
20	11059	1998-04-29	RICAR	Ricardo Adockados
21	11057	1998-04-29	NORTS	North/South
22	11054	1998-04-28	CACTU	Cactus Comidas para llevar
23	11056	1998-04-28	EASTC	Eastern Connection
24	11055	1998-04-28	HILAA	HILARIOUX-Bistro
25	11051	1998-04-27	LAMAS	La Maison d'Asie
26	11045	1998-04-17	ANNO	Anno's and Sons

Showing rows: 1 to 830 Page No: 1 of 1

Total rows: 830 Query complete 00:00:00.082

MS 443 - In-term T... pgAdmin 4

9:57 AM 4/08/2025

Q4

pgAdmin 4

Object Explorer

- Publications
- Schemas (2)
 - exam
 - Aggregates
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (1)
 - Columns
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - Trigger Functions
 - Types
 - Views
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (14)
 - categories
 - customer_customer_demo
 - customer_demographics
 - customers
 - employee_territories
 - employees

Query

```

-- Question 4: Northwind management wants to identify large product movements to better plan inventory and logistics. Write an SQL query to display orders where a product was purchased in large quantity (no
-- Your answer here
select od.order_id, p.product_id, p.product_name, od.quantity
from order_details od
join products p
on od.product_id = p.product_id
where od.quantity > 99
order by od.quantity desc;
-- Question 5: Northwind management wants to evaluate the delivery performance of each shipping partner. Write an SQL query to calculate the average delivery time (in days) for each shipper. Delivery time =
-- Your answer here

```

Data Output

order_id	product_id	product_name	quantity
1	10702	Wimmers gute Semmelbrödel	130
2	10764	Charmouse verte	130
3	10451	Pâté chinois	120
4	10919	Sauage Schokolade	120
5	10711	Petit Pâtisier	120
6	10999	Pâté chinois	120
7	10593	Sirop d'érable	120
8	10678	Jack's New England Clam Chow...	120
9	10776	Maryknig-Süßes Apfels	120
10	10864	Rheinische Kirschtorte	120
11	11017	Raspélie Courdévault	110
12	10713	Rogede wild	110
13	10893	Guaraní Fantástica	110
14	11030	Raspélie Courdévault	100
15	10654	Sauage	100
16	10453	Gula Molesta	100
17	10893	Camembert Pâtisier	100
18	10588	Singaporean Hokkien Fried Mee	100
19	10607	Alize Mutton	100
20	11030	Champ	100
21	10678	Quatre Manhegin La Pastora	100
22	10266	Swalaya Stout	100
23	10569	Rogede wild	100

Total rows: 23 Query complete 00:00:00.058

Q5

pgAdmin 4

Object Explorer

- Publications
- Schemas (2)
 - exam
 - Aggregates
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (14)
 - categories
 - customer_customer_demo
 - customer_demographics
 - customers
 - employee_territories
 - employees
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (14)
 - categories
 - customer_customer_demo
 - customer_demographics
 - customers
 - employee_territories
 - employees

Query

```

-- Question 5: Northwind management wants to evaluate the delivery performance of each shipping partner. Write an SQL query to calculate the average delivery time (in days) for each shipper. Delivery time =
-- Your answer here
select s.shipper_id, s.company_name,
avg(o.shipped_date - o.order_date) as avg_delivery_days
from shippers s
join orders o
on s.shipper_id = o.ship_id
where o.shipped_date is not null
group by s.shipper_id, s.company_name;
-- Question 6: Northwind wants to identify the most active customers (customers who place orders most frequently) to target retention campaigns. Write an SQL query to rank customers based on their total num
-- Your answer here

```

Data Output

shipper_id	company_name	avg_delivery_days
1	Speedy Express	6.5714285714285714
2	Federal Shipping	7.47389532329173
3	United Package	8.2348208348208349

Total rows: 3 Query complete 00:00:00.063

Q6

